



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2018 – 0458



Agricultural Machinery – Forage Chopper CIAMC 9 ZP-0.4

Test Report Number	2018 – 0458
Date of Issuance	August 8, 2019
Valid Until	August 8, 2024
Agricultural Machinery	Forage Chopper
Type	Four-blade, Mobile type, Power-operated
Brand	CIAMC
Model	9 ZP-0.4
Test Requested by	CENTRAL ISABELA AGRI MANUFACTURING CORPORATION, Tagaran, Cauayan City, Isabela

SUMMARY

Performance Criteria	Requirement as per PAES 218:2004	AMTEC Test
Variation of Cut, %, maximum	5	20.6
Clear Cut Chopped Material	Yes	Yes
Noise Level, maximum, dB(A), maximum	96*	99.9

Note: * Allowable noise level for four (4) hours of continuous exposure based on Occupational Safety and Health Standards, Ministry of Labor, Philippines. 1983.

OBSERVATIONS

1. Operator's manual	Not provided
2. Previous valid test reports of similar brand and model	None
3. Manufacturer	CENTRAL ISABELA AGRI MANUFACTURING CORPORATION
4. Serial number	ND
5. Prime mover	4.8 kW TOYOHAMA TH168F2 Gasoline Engine
6. Cutter head knives	Adjustable and replaceable.
7. Adjusting and repair of parts	Not observed
8. Mechanism for easy disengagement of the machine from the engine	A clutch mechanism was provided.
9. Transporting the machine	The machine has rubber wheels for transport.
10. Safety	Transmission system was provided with cover.
11. Number of operator	1 Feeding operator, 1 Optional bagger
12. Breakdown/Failure	No breakdown occurred.

13. Other remarks	The machine was provided with a feeding conveyor.
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PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	May 10, 2018
Location of Test	Tagaran, Cauayan City, Isabela
Items Tested	Output capacity, Cutting efficiency, Output quality, Speed of components, Fuel consumption, Noise level, and Number of operators

METHODS OF TEST

The forage chopper was tested based on the PAES 219:2004 Agricultural Machinery – Forage Chopper – Methods of Test.

DESCRIPTION OF THE MACHINE

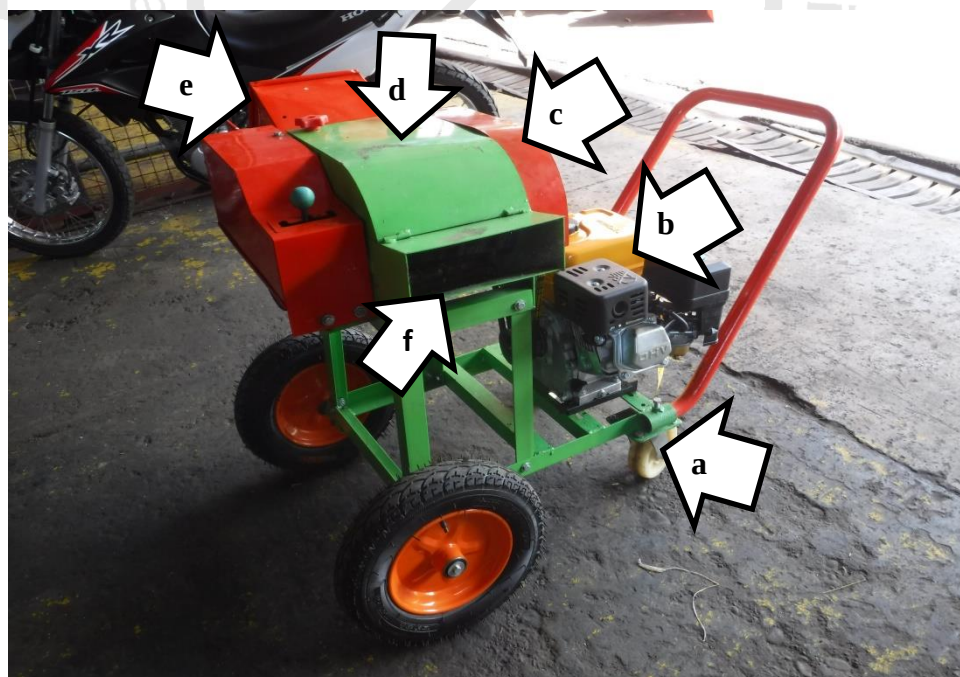


Figure 1. Parts of Forage Chopper CIAMC 9 ZP-0.4 powered 4.8 kW TOYOHAMA TH168F2 Gasoline Engine include the following: (a) Transport wheels, (b) Prime mover, (c) Belt-pulley transmission system, (d) Cutter assembly, (e) Input chute, and (f) Discharge chute.

SPECIFICATIONS

Item		Manufacturer's Specification ^a	AMTEC Verification ^b
1 Overall Dimensions, mm			
1.1	Length	ND	1135
1.2	Width	ND	1150
1.3	Height	ND	825
1.4	Weight, without primemover, kg	ND	NM
2 Cutting Blades			
2.1	Shape	ND	Rectangular
2.2	Dimension, L x W x t, mm	ND	220 x 70 x 16
2.3	Number of cutter knives	ND	4
2.4	Materials	ND	Tool steel
3 Feeding Table			
3.1	Dimension, L x W, mm	ND	225 x 75
3.2	Height from the ground, mm	ND	655
3.3	Feeding conveyor size, L x W, mm	ND	500 x 220
3.4	Feeder roller, L x D, mm	ND	220 x 54
4 Output Chute,			
4.1	Dimension, L x D, mm	ND	230 x 80
4.2	Height from the ground	ND	630
5 Main Frame			
5.1	Material	ND	Angle Bar
5.2	Size, L x W x T, mm	ND	38 x 38 x 4
6 Transmission System			
6.1	Engine to chopper	ND	A-belt and pulley
6.2	Engine pulley	ND	76 mm x 2A x 19 mm
6.3	Chopper pulley	ND	190 mm x 2A x 24 mm
6.4	Belt size	ND	A – 37

Note: ^a Data obtained from the nameplate/stickers on the machine
^b Data obtained from the actual test and on site measurement.
 ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification ^b
7	Prime Mover		
7.1	Engine		
7.1.1	Brand	ND	TOYOHAMA
7.1.2	Model	ND	TH168F2
7.1.3	Make	ND	ND
7.1.4	Serial number	ND	ND
7.1.5	Type	ND	Four-stroke cycle, OHV, single cylinder, gasoline engine
7.1.6	Rated power, kW	ND	4.8
7.1.7	Rated speed, rpm	ND	3600
7.1.8	Cooling system	ND	Air-cooled
7.1.9	Starting system	ND	Rope recoil starter
7.1.10	Weight, kg	ND	NM
8	Front Wheels		
8.1	Type	ND	Rubber tires (fixed)
8.2	Size, D x W, mm	ND	360 x 75
8.3	Number of pieces	ND	2
9	Rear Wheels		
9.1	Type	ND	Rotating synthetic rubber
9.2	Size, D x W, mm	ND	130 x 40
9.3	Number of pieces	ND	2

Note: ^a Data obtained from the nameplate/stickers on the machine

^b Data obtained from the actual test and on site measurement.

ND – No Data, NM – Not Measured

RESULTS

Item		Data	
1	Conditions of Test Material		
1.1	Type	Corn plants	
1.2	Variety	ND	
1.3	Source	Unknown	
1.4	Maturity, days	120	
1.5	Moisture content, % _{wb}	58.80	
1.6	Size of cut plants, L x D, mm	NM	
2	Performance Test		
2.1	Weight of output, kg	26.1	
2.2	Output time, h	0.0465	
2.3	Output capacity, kg/h	561.3	
2.4	Chopping efficiency, %	100	
2.5	Average length of cut, mm	19	
2.6	Variation of cut, %	20.6	
2.7	Speed of components, rpm	Without Load	With Load
2.7.1	Engine	3868	3649
2.7.2	Chopper blade	1337	1284
2.8	Noise level, dB(A)	Without Load	With Load
2.8.1	Feeding operator	93.5	96.7
2.8.2	Bagger	96.5	99.9
2.9	Fuel consumed, mL	85	
2.10	Fuel time, h	0.0702	
2.11	Fuel consumption, L/h	1.21	

OBSERVATIONS

Figure 2. Quality of output.

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AUG 08 2019
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